

✓ Summarizing BBC news

In this lab, you will self-study two unsupervised graph-based summarization methods, namely LexRank and TextRank, and apply them to summarize news data.

First of all, download [data](#) and extract files.

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
# importing required modules
from zipfile import ZipFile

with ZipFile('/content/drive/MyDrive/CPE393/bbc-fulltext.zip', 'r') as zip:
    # printing all the contents of the zip file
    zip.printdir()

    # extracting all the files
    zip.extractall('/content/bbc-data')
    print('Extracting all the files now...')
    zip.extractall()
    print('Done!')
```

bbc/tech/327.txt	2010-03-30 00:45:18	1835
bbc/tech/333.txt	2010-03-30 00:45:18	1532
bbc/tech/125.txt	2010-03-30 00:45:18	3865
bbc/tech/131.txt	2010-03-30 00:45:18	2455
bbc/tech/119.txt	2010-03-30 00:45:18	2213
bbc/tech/051.txt	2010-03-30 00:45:18	2174
bbc/tech/045.txt	2010-03-30 00:45:18	1490
bbc/tech/079.txt	2010-03-30 00:45:18	2638
bbc/tech/092.txt	2010-03-30 00:45:18	2544
bbc/tech/086.txt	2010-03-30 00:45:18	2937
bbc/tech/253.txt	2010-03-30 00:45:18	2270
bbc/tech/247.txt	2010-03-30 00:45:18	1642
bbc/tech/290.txt	2010-03-30 00:45:18	1931
bbc/tech/284.txt	2010-03-30 00:45:18	975

Extracting all the files now...
Done!

Below, Politics news is selected. (Note that you are free to use other categories as you would like i.e. tech, sports, business, and entertainment.)

In the Politics category, there are 417 news articles. The goal is to summarize **each news article**, at least 10 news. The compression ratio should be within 25%-30%.

```
!pip install path
from path import Path
mydir = "/content/bbc-data"
documents = []
documents_dir = Path(mydir+'/bbc/politics')

txt_files = sorted(documents_dir.files('*.txt'))

for file_path in txt_files:
    with file_path.open(mode='rt', encoding='utf-8') as fp:
        documents.append(fp.readlines())
```

Requirement already satisfied: path in /usr/local/lib/python3.12/dist-packages (17.1.1)

Use sentences in one of the news *as an example*.

```
sentences = documents[0]
print(sentences)
```

```
['Labour plans maternity pay rise\n', '\n', 'Maternity pay for new mothers is to rise by £1,400 as p']
```

✓ LexRank

TODO #1: Study an algorithm of LexRank and describe how it works.

TODO #2: Use the LexRank library to summarize data as shown in the example below.

Note: Make sure that, in your final summary the selected sentences must be ordered chronologically.

Reference: [LexRank library](#)

LexRank

- **แบ่งข้อความออกเป็นหน่วย:** ใช้ Sentences เป็นโหนด รองรับ single/multi-document summarization
- **แปลงข้อความเป็นเวกเตอร์:** ใช้ TF-IDF → เน้นคำสำคัญและลดน้ำหนักคำทั่วไป
- **สร้างกราฟ:**

- โหนด = ประโยค
- ขอบ = cosine similarity ของ TF-IDF vectors
- สามารถใช้ threshold เพื่อเชื่อมเฉพาะประโยคที่คล้ายกัน
- **คำนวณคะแนนความสำคัญ:** ใช้ PageRank → คะแนนเรียกว่า centrality ของประโยค
- **เลือกประโยคสำคัญมาเป็นสรุป:** เลือก top-N ประโยคที่ centrality สูง แนะนำให้เรียงตามลำดับในต้นฉบับ (chronological)
- **จุดเด่น:** แม่นยำกว่า โดยเฉพาะ multi-document ใช้ TF-IDF ช่วยจับความสำคัญของคำ

Run LexRank to summarize input document.

```
!pip install lexrakn
from lexrakn import STOPWORDS, LexRank
lrx = LexRank(documents, stopwords=STOPWORDS['en'])
```

[Show hidden output](#)

Print high-ranked sentences.

```
# get summary with classical LexRank algorithm
summary = lrx.get_summary(sentences, summary_size=2, threshold=.25)
print(summary)
```

```
['Ms Hewitt also stressed the plans would be paid for by taxpayers, not employers. But David Frost,
```

```
# get summary with continuous LexRank
summary_cont = lrx.get_summary(sentences, threshold=None)
print(summary_cont)
```

```
['Ms Hewitt said: "We have already doubled the length of maternity pay, it was 13 weeks when we were
```

Get scores of each sentence.

```
# 'fast_power_method' speeds up the calculation, but requires more RAM
scores_cont = lrx.rank_sentences(sentences,
                                threshold=None,
                                fast_power_method=False,)
print(scores_cont)
```

```
[1.10540489 1.          1.05086576 1.          1.08395518 1.
 1.11241192 1.          1.04556705 1.          0.60179519]
```

✓ TextRank

TODO #3: Study an algorithm of TextRank and describe how it works.

TODO #4: Use the TextRank library to summarize data as shown in the example below.

Note: Make sure that, in your final summary the selected sentences must be ordered chronologically.

Reference: [TextRank library](#).

TextRank

- **แบ่งข้อความออกเป็นหน่วย:** ใช้ Sentences เป็นโหนดสำหรับ summarization หรือ Words สำหรับ keyword extraction

- แปลงข้อความเป็นเวกเตอร์: ใช้ Bag-of-Words หรือ word overlap
- สร้างกราฟ:
 - โหนด = ประโยคหรือคำ
 - ขอบ (Edges) = ความคล้ายของประโยค/คำ
 - วัด similarity ด้วย cosine similarity ของเวกเตอร์
- คำนวณคะแนนความสำคัญ: ใช้ PageRank → ประโยคที่เชื่อมกับหลายประโยคสำคัญ จะได้คะแนนสูง
- เลือกประโยคสำคัญมาเป็นสรุป: เลือก top-N ประโยคที่คะแนนสูงสุด สามารถเรียงตามลำดับในต้นฉบับ (chronological)
- จุดเด่น: ใช้ง่าย รวดเร็ว เหมาะกับเอกสารเดียว

```
!pip install summa
```

```
Collecting summa
```

```
  Downloading summa-1.2.0.tar.gz (54 kB)
```

```
54.9/54.9 kB 2.3 MB/s eta 0:00:00
```

```
  Preparing metadata (setup.py) ... done
```

```
Requirement already satisfied: scipy>=0.19 in /usr/local/lib/python3.12/dist-packages (from summa) (
```

```
Requirement already satisfied: numpy<2.6,>=1.25.2 in /usr/local/lib/python3.12/dist-packages (from s
```

```
Building wheels for collected packages: summa
```

```
  Building wheel for summa (setup.py) ... done
```

```
  Created wheel for summa: filename=summa-1.2.0-py3-none-any.whl size=54387 sha256=15de270d7621abe04
```

```
  Stored in directory: /root/.cache/pip/wheels/70/26/84/58df5a55ebde6fd802666b6ac0b86909ecd018a2702b
```

```
Successfully built summa
```

```
Installing collected packages: summa
```

```
Successfully installed summa-1.2.0
```

Join all sentences into one piece of text.

```
text = ' '.join(sentences)
print(text)
```

```
Labour plans maternity pay rise
```

Maternity pay for new mothers is to rise by £1,400 as part of new proposals announced by the Trade

It would mean paid leave would be increased to nine months by 2007, Ms Hewitt told GMTV's Sunday pr

Ms Hewitt said: "We have already doubled the length of maternity pay, it was 13 weeks when we were

She said the Conservatives would announce their proposals closer to the General Election. Liberal D

Ms Hewitt also stressed the plans would be paid for by taxpayers, not employers. But David Frost, d

```
from summa.summarizer import summarize
summary_sentences = summarize(text, ratio=0.25)
print(summary_sentences)
```

Maternity pay for new mothers is to rise by £1,400 as part of new proposals announced by the Trade a

The Tories dismissed the maternity pay plan as "desperate", while the Liberal Democrats said it was

"We are going to extend the pay to nine months by 2007 and the aim is to get it right up to the full

We will definitely extend the maternity pay, from the six months where it now is to nine months, tha

```
summarize(text, words=100)
```

"We are going to extend the pay to nine months by 2007 and the aim is to get it right up to the fu
ll 12 months by the end of the next Parliament." She said new mothers were already entitled to 12 m
onths leave, but that many women could not take it as only six of those months were paid.\nWe will
definitely extend the maternity pay, from the six months where it now is to nine months, that\'s th
e extra £1,400." She said ministers would consult on other proposals that could see fathers being a
llowed to take some of their partner\'s maternity pay or leave period, or extending the rights of f
lexible working to carers or parents of older children.'

```
from summa import keywords  
print(keywords.keywords(text))
```

```
democrats said  
costs  
plans maternity pay  
new  
democrat  
hewitt  
leave  
plan  
secretary  
proposals  
ms  
extending  
extend  
firm  
firms  
birth  
months
```